

Algebra | equations and systems

1. $19y - 13y + 7y - 1 = 4 - 14 - 12 + 2y + 20 - 19y - 11$

2. $3x^2 - 9x = 0$

3. $3x^3 + 6x^2 - 27x - 54 = 0$

4. Solve the system of equations:

$$6x + 3x - 4y + 8y + 8 = 3x + 8y + 2$$

$$-8x + 32y = 88$$

Solutions

1. -0.4

2. $0, 3$

3. $-2, -3, 3$

4. $x = 1, y = 3$

Functions | from expression to graph

$$x^3 - 2x + 1$$

5. The function $f(x) = -2x^2 + 2x + 24$ is given.

- a) What is the domain of the function?
- b) What is the derivative of the function?
- c) What are the extremums of the function?
- d) When is the function increasing, and when is it decreasing?
- e) Find the horizontal asymptotes of the function (if there are any).
- f) sketch the function.

Solutions

5. a. Domain: all real numbers

b. Derivative: $-4x + 2$

c. Extremums: (0.5, 24.5)

d. Increase & Decrease: Increase: $x < 0.5$, Decrease: $0.5 < x$

e. Horizontal Asymptotes: None (nonconstant polynomial).

f. See the sketch below.

